Supplementary Information

Paired Fault Assessment of a Fly-Connectome Spiking Controller

Shuchao Gao, Waseda University

Assessment v9: matched source, data and protocol

This document separates the independent core test, a prespecified angle stress test, development, and historical audits. Every new table is reconstructed from saved raw records and frozen registrations. Historical observations are retained with their original limitations; they are not silently relabeled as new experiments.

1 Protocol, software and calibration

The current task is `task-v9.1-causal-exact-events`. The main Methods and `paper/PROTOCOL_v9.md` define the sampled plant, common-random-number coupling, first-passage task outcome and resource estimands. All test parameters were frozen before outcomes. The neural substrate uses released FlyWire v783 tables and Shiu parameters; the original publication used v630. The synthetic visual/body coupling is not validated fly behavior. Functional row deletions are not physical bit faults. Between spikes the voltage and voltage-valued synaptic drive obey

$$\dot{v}_i = (v_0 - v_i + g_i)/(20\,\mathrm{ms}), \qquad \dot{g}_i = -g_i/(5\,\mathrm{ms}).$$

Recurrent events add signed contact counts times 0.275 mV to g_i after 1.8 ms. At threshold -45 mV, reset sets v_i to $v_0 = -52$ mV and g_i to zero; both equations pause during 2.2 ms refractoriness. Sensory cells retain this refractoriness, a variant of upstream stimulation. External events add 68.75 mV to voltage; all non-sensory external rates are zero. Initial neural states, monitors and unmodified recurrent weights are restored before each episode. A separate tiny actual Brian2 test verifies this reset, including reapplying external weights, without loading a full connectome.

Table S1: Fixed physical settings. No hardware power or area is inferred.

Parameter	Value
Neural / feedback step	0.1 / 20 ms
Cue / task horizon	100 / 1000 ms
Pursuer speed / initial separation	0.3 m/s / 0.25 m
Contact / looming radius	0.05 / 0.05 m
Walk / burst speed	0.05 / 0.35 m/s
Burst delay / duration	250 / 450 ms
Common turn limit / heading cap	600 deg/s / ±100 deg
RF seed / inactive pool seed	20260201 / 0

Independent input streams use SeedSequence [20261022, seed, frame, 1]; deletion uses [20261022, seed, 2]. The former supplies float32 cell/time uniforms and the latter float64 row uniforms. Both doses reuse the same row variates. Actual realized contact mass and sign composition are saved, not forced to equal their expectations. The input map saves pool/sensory/command identities and RF parameters. Exact source and table hashes are bound by the JSON registrations.

Table S2: Fitted direction decoder coefficients, raw five-frame Hz features. Intercept is 0.026677472.

FlyWire ID	Type	Side	β_i
720575940622838154	DNp01	left	-0.005881965
720575940632499757	DNp01	right	-0.004112180
720575940604737708	DNa02	right	0.000000000
720575940629327659	DNa02	left	0.000000000
720575940604954289	DNp04	left	0.012368436
720575940629757036	DNp04	right	-0.005714159
720575940625878160	DNb01	left	0.034961485
720575940630758418	DNb01	right	0.000000000

Development fits bearing/60° with standardized inputs, ridge penalty 10 and an unpenalized intercept over 200–400 ms. The burst threshold is half the tenth percentile of per-episode 200–300 ms mean command rate, floor 2.5 Hz; here it is 4.03125 Hz. Development fit quality is not held-out prediction accuracy. The published B gain/silence gate followed a failed developmental attempt; all attempts remain available.

Table S3: Unique newly executed records by role; no reused healthy test records.

Role	Seeds	Episodes	Horizon	Outcome use
Passive development	200–203	20	400 ms	Fit only
Stopped A configuration	300–302	6	1000 ms	Development
Healthy B validation	400–407	32	1000 ms	32/32 survived
B fault screen	400,401	16	1000 ms	15/16 survived
Independent core	1000–1023	288	1000 ms	24 clusters
Angle stress	1100–1107	96	1000 ms	8 separate clusters
Action ablations	1200–1207	64	1000 ms	8 separate clusters

The initial A configuration stopped after healthy success was 1/3, making its predeclared 3/4-per-stratum screen impossible. Its three F2_25 records included two survivors. B changes the common gain to 600 deg/s and gates steering during silence, preserving the fitted decoder/trigger. All 32 B healthy validations survived (8/8 in every law/bearing stratum). The two-seed fault screen retained a proportional/-30/F2_50 collision at 977.0661 ms. These are developmental observations, not independent confirmation.

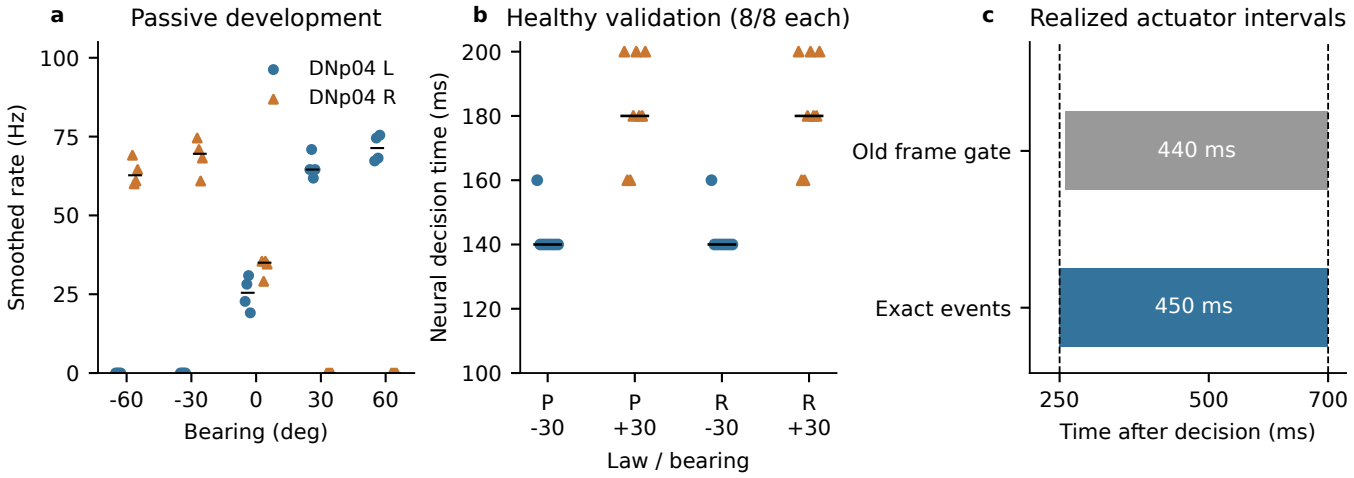


Figure S1: Development and validation diagnostics, not independent-test evidence. (a) DNp04 mean smoothed rates in 200–400 ms; all four developmental seeds per bearing, slight horizontal jitter for visibility, black median. (b) All eight validation seeds per law/bearing; P/R denote proportional/relay, black median. The same seed is shared across conditions. (c) Observed old completed bursts (four v8 records) and all 32 complete B-validation bursts, relative to decision time. This timing comparison does not isolate the cause of old-versus-new task outcomes; other controller/protocol components also changed.

2 Independent core outcomes and seed clusters

Table S4: Core outcomes. Each direction/law/dose has 24 seed clusters. L/G: healthy-to-fault losses/gains; H: healthy survivors at risk. Healthy rows compare to themselves.

Law	Network	ϕ	Success	L	G	H at risk
proportional	healthy	-30	24/24	0	0	24
proportional	healthy	+30	24/24	0	0	24
proportional	F2_25	-30	23/24	1	0	24
proportional	F2_25	+30	23/24	1	0	24
proportional	F2_50	-30	14/24	10	0	24
proportional	F2_50	+30	23/24	1	0	24
relay	healthy	-30	24/24	0	0	24
relay	healthy	+30	24/24	0	0	24
relay	F2_25	-30	24/24	0	0	24
relay	F2_25	+30	24/24	0	0	24
relay	F2_50	-30	24/24	0	0	24
relay	F2_50	+30	24/24	0	0	24

Table S5: Every core task outcome, 1=collision-free through 1000 ms. P/R: proportional/relay. H/25/50: healthy and row deletion percentages. No direction or failed seed is omitted.

Seed	ϕ	P H	P 25	P 50	R H	R 25	R 50
1000	-30	1	1	1	1	1	1
1000	+30	1	1	1	1	1	1
1001	-30	1	1	1	1	1	1
1001	+30	1	1	1	1	1	1
1002	-30	1	1	0	1	1	1
1002	+30	1	1	1	1	1	1
1003	-30	1	1	0	1	1	1
1003	+30	1	1	1	1	1	1
1004	-30	1	1	1	1	1	1
1004	+30	1	1	1	1	1	1
1005	-30	1	1	0	1	1	1
1005	+30	1	0	1	1	1	1
1006	-30	1	1	1	1	1	1
1006	+30	1	1	0	1	1	1
1007	-30	1	0	1	1	1	1
1007	+30	1	1	1	1	1	1
1008	-30	1	1	0	1	1	1
1008	+30	1	1	1	1	1	1
1009	-30	1	1	1	1	1	1
1009	+30	1	1	1	1	1	1
1010	-30	1	1	0	1	1	1
1010	+30	1	1	1	1	1	1
1011	-30	1	1	0	1	1	1
1011	+30	1	1	1	1	1	1
1012	-30	1	1	0	1	1	1
1012	+30	1	1	1	1	1	1
1013	-30	1	1	1	1	1	1
1013	+30	1	1	1	1	1	1
1014	-30	1	1	1	1	1	1
1014	+30	1	1	1	1	1	1
1015	-30	1	1	1	1	1	1
1015	+30	1	1	1	1	1	1
1016	-30	1	1	1	1	1	1
1016	+30	1	1	1	1	1	1
1017	-30	1	1	0	1	1	1
1017	+30	1	1	1	1	1	1

Seed	ϕ	P H	P 25	P 50	R H	R 25	R 50
1018	-30	1	1	1	1	1	1
1018	+30	1	1	1	1	1	1
1019	-30	1	1	1	1	1	1
1019	+30	1	1	1	1	1	1
1020	-30	1	1	0	1	1	1
1020	+30	1	1	1	1	1	1
1021	-30	1	1	1	1	1	1
1021	+30	1	1	1	1	1	1
1022	-30	1	1	0	1	1	1
1022	+30	1	1	1	1	1	1
1023	-30	1	1	1	1	1	1
1023	+30	1	1	1	1	1	1

Table S6: Complete-seed effects in percentage points; 20,000 percentile bootstrap resamples, seed 20261022, conditional on this model/map/task.

Seed-level estimand	Mean (pp)	95% interval (pp)
proportional_F2_25_net_loss	4.2	[0.0, 10.4]
proportional_F2_50_net_loss	22.9	[12.5, 33.3]
relay_F2_25_net_loss	0.0	[0.0, 0.0]
relay_F2_50_net_loss	0.0	[0.0, 0.0]
F2_25_interaction	-4.2	[-10.4, 0.0]
F2_50_interaction	-22.9	[-33.3, -12.5]

Table S7: Every actual independent cluster contrast. P/R and 25/50 match the preceding table; interaction is R minus P net loss.

Seed	P25	R25	Δ25	P50	R50	Δ50
1000	0.00	0.00	0.00	0.00	0.00	0.00
1001	0.00	0.00	0.00	0.00	0.00	0.00
1002	0.00	0.00	0.00	0.50	0.00	-0.50
1003	0.00	0.00	0.00	0.50	0.00	-0.50
1004	0.00	0.00	0.00	0.00	0.00	0.00
1005	0.50	0.00	-0.50	0.50	0.00	-0.50
1006	0.00	0.00	0.00	0.50	0.00	-0.50
1007	0.50	0.00	-0.50	0.00	0.00	0.00
1008	0.00	0.00	0.00	0.50	0.00	-0.50
1009	0.00	0.00	0.00	0.00	0.00	0.00
1010	0.00	0.00	0.00	0.50	0.00	-0.50
1011	0.00	0.00	0.00	0.50	0.00	-0.50
1012	0.00	0.00	0.00	0.50	0.00	-0.50
1013	0.00	0.00	0.00	0.00	0.00	0.00
1014	0.00	0.00	0.00	0.00	0.00	0.00
1015	0.00	0.00	0.00	0.00	0.00	0.00
1016	0.00	0.00	0.00	0.00	0.00	0.00
1017	0.00	0.00	0.00	0.50	0.00	-0.50
1018	0.00	0.00	0.00	0.00	0.00	0.00
1019	0.00	0.00	0.00	0.00	0.00	0.00
1020	0.00	0.00	0.00	0.50	0.00	-0.50
1021	0.00	0.00	0.00	0.00	0.00	0.00
1022	0.00	0.00	0.00	0.50	0.00	-0.50
1023	0.00	0.00	0.00	0.00	0.00	0.00

Table S8: Prespecified symmetric-null sign-flip sensitivity; not a randomized-controller assignment test.

Dose	Raw p	Holm p
F2_25	0.500000	0.500000
F2_50	0.000977	0.001953

The originally frozen analyzer passed the complete input/source/timing checks but then failed on duplicate numeric-margin/Boolean-warning CSV keys, before reporting statistics. The original script, failure log and output directory are retained. The documented exportfix analyzer renames only numeric CSV fields, preserving all numerical definitions, thresholds, source guards and raw records; canonical analyses are core-v2 and stress-v2. This is a post-registration analytical export correction, not a new hypothesis or neural rerun. See `paper/PROTOCOL_v9_NOTES.md`.

A post-hoc control diagnostic verifies identical first trigger times and pre-decision traces in all 144 paired law/scenario cells. Both laws gate steering below the trigger threshold, so their first action divergence follows that same decision. Table entries at 400 ms precede every contact. Equal maximum turn rate does not match executed turning effort; these diagnostics are not causal mediation estimates or energy measurements.

Table S9: Descriptive precontact control diagnostics on all core records; medians pool the two paired directions, without adding independent samples.

Law	Network	Median trigger (ms)	Median $\lvert\theta(400)\rvert$ (deg)
proportional	healthy	160.0	88.27
proportional	F2_25	180.0	71.62
proportional	F2_50	220.0	38.73
relay	healthy	160.0	100.00
relay	F2_25	180.0	100.00
relay	F2_50	220.0	96.00

Table S10: Healthy baseline safety reserve over the full 1-s episode. No healthy record collided, so these values contain no post-contact continuation. Matching binary success does not match clearance or executed effort.

Law	Episodes	Median clearance (mm)	Minimum (mm)	Maximum (mm)
proportional	48	33.08	13.88	39.64
relay	48	41.43	39.24	42.19

3 Frozen warnings, information and error denominators

Table S11: Thresholds learned only from all 32 healthy validation records.

Criterion	Threshold
Command mean, 100–400 ms	5.833333333 Hz
Two-sided activity log score	0.169456319
Held-command clearance	0.018772702 m
Schedule-only clearance	0.018548260 m
Upper resource excursion	2× reference

All prospective warnings use data through 400 ms, verified before first contact in every record. The two-sided activity score is the absolute log count ratio to the law/side-specific validation median. Its cutoff is the healthy 95th percentile; command and forecast cutoffs are fifth percentiles. The same nominal 5% healthy flag calibration is not a guarantee of task FPR. Held-command and schedule forecasts share exact simulated geometry and the pending actuator schedule; the candidate holds the current neural-derived turn command, the baseline sets future turn to zero. If no trigger exists, both assume the earliest next-frame decision. This is an optimistic extrapolation, not a neural-only detector or reachability proof.

Table S12: Core confusion counts. Positives are collisions through 1 s. The own-stop upper criterion is retrospective resource scoring, not a 400-ms warning. All/faulty/healthy subsets are overlapping views of the same seed clusters. A dash means the conditional rate is undefined because its class is absent.

Criterion	Subset	TP	FP	TN	FN	Sens	Spec	FPR	FNR
Command low	all	2	51	224	11	0.154	0.815	0.185	0.846
Command low	faulty	2	49	130	11	0.154	0.726	0.274	0.846
Command low	healthy	0	2	94	0	–	0.979	0.021	–
Own-stop upper	all	3	8	267	10	0.231	0.971	0.029	0.769
Own-stop upper	faulty	3	8	171	10	0.231	0.955	0.045	0.769
Own-stop upper	healthy	0	0	96	0	–	1.000	0.000	–
400-ms upper	all	0	0	275	13	0.000	1.000	0.000	1.000
400-ms upper	faulty	0	0	179	13	0.000	1.000	0.000	1.000
400-ms upper	healthy	0	0	96	0	–	1.000	0.000	–
Two-sided activity	all	3	68	207	10	0.231	0.753	0.247	0.769
Two-sided activity	faulty	3	54	125	10	0.231	0.698	0.302	0.769
Two-sided activity	healthy	0	14	82	0	–	0.854	0.146	–
Actuator schedule	all	13	84	191	0	1.000	0.695	0.305	0.000
Actuator schedule	faulty	13	80	99	0	1.000	0.553	0.447	0.000
Actuator schedule	healthy	0	4	92	0	–	0.958	0.042	–
Held-command candidate	all	13	57	218	0	1.000	0.793	0.207	0.000
Held-command candidate	faulty	13	55	124	0	1.000	0.693	0.307	0.000
Held-command candidate	healthy	0	2	94	0	–	0.979	0.021	–

The retrospective own-stop upper flag divides by its paired healthy stopped count. The prospective 400-ms activity flags instead use frozen law/side validation medians. Their classification difference therefore changes both observation and reference, and is not a pure exposure ablation. The duration-isolated comparison is R_{stop} versus R_{common} within the same healthy/fault pair. The two-sided score also detects low activity; it addresses a different hypothesis from an upper resource limit.

Table S13: Secondary descriptive paired held-command minus schedule comparison (pp), added after registration. The same bootstrap draws keep complete seed clusters together. No extra confirmatory test or threshold tuning is introduced.

Subset	Metric	Difference (pp)	95% interval (pp)
all	sensitivity	0.00	[0.00, 0.00]
all	specificity	9.82	[6.57, 13.41]
all	error_rate	-9.38	[-12.85, -6.25]
faulty	sensitivity	0.00	[0.00, 0.00]
faulty	specificity	13.97	[9.44, 19.02]
faulty	error_rate	-13.02	[-17.71, -8.85]
healthy	sensitivity	–	–
healthy	specificity	2.08	[0.00, 5.21]
healthy	error_rate	-2.08	[-5.21, 0.00]

Table S14: Sparse-class bootstrap diagnostics. A missing positive class makes sensitivity undefined; such draws are not silently treated as zero.

Subset	Positive clusters	Negative clusters	Valid sens. draws	Valid spec. draws
all	12	24	20000	20000
faulty	12	24	20000	20000
healthy	0	24	0	20000

Table S15: Direction/law disaggregation of the same core warning records.

Criterion	Stratum	TP	FP	TN	FN	Sens	Spec
transmission	proportional/-30	1	0	61	10	0.091	1.000
transmission	proportional/30	1	12	58	1	0.500	0.829
transmission	relay/-30	0	13	59	0	–	0.819
transmission	relay/30	0	26	46	0	–	0.639
equal_duration_upper	proportional/-30	0	0	61	11	0.000	1.000
equal_duration_upper	proportional/30	0	0	70	2	0.000	1.000
equal_duration_upper	relay/-30	0	0	72	0	–	1.000
equal_duration_upper	relay/30	0	0	72	0	–	1.000
equal_duration_two_sided	proportional/-30	2	12	49	9	0.182	0.803
equal_duration_two_sided	proportional/30	1	8	62	1	0.500	0.886
equal_duration_two_sided	relay/-30	0	25	47	0	–	0.653
equal_duration_two_sided	relay/30	0	23	49	0	–	0.681
schedule_only	proportional/-30	11	31	30	0	1.000	0.492
schedule_only	proportional/30	2	37	33	0	1.000	0.471
schedule_only	relay/-30	0	0	72	0	–	1.000
schedule_only	relay/30	0	16	56	0	–	0.778
deadline	proportional/-30	11	27	34	0	1.000	0.557
deadline	proportional/30	2	21	49	0	1.000	0.700
deadline	relay/-30	0	0	72	0	–	1.000
deadline	relay/30	0	9	63	0	–	0.875

Full conditional cluster intervals for sensitivity, specificity, FPR and FNR are in the source summary JSON. Resampling keeps every law/bearing/dose within seed together. Zero observed errors may produce a degenerate bootstrap interval; the one-sided 95% bound for the probability of at least one error in an otherwise identically defined seed cluster is $1 - 0.05^{1/n}$. This is a cluster-any-error bound, not a conditional FPR/FNR guarantee. No population or hardware zero-risk claim follows.

The original registration gives an approximate 0.20 planning half-width without stating its range explicitly. This normal maximum-variance approximation applies only to $[0, 1]$ success/loss/gain cluster means. Signed net loss spans $[-1, 1]$, and its between-law contrast can span $[-2, 2]$, yielding range-based planning approximations about 0.40 and 0.80, respectively. These are not finite-sample coverage guarantees. The clarification in `paper/PROTOCOL_v9_NOTES.md` preserves the registration bytes, sample size and analysis. Actual intervals are reported; small interactions and equivalence are not established by the sample size.

4 Exposure and observation horizon

Table S16: Paired network-spike estimands. Own-stop and common-duration use recorded first-contact frame ends. Fixed 1-s activity includes explicitly simulated post-contact continuation.

Law/dose	Estimand	Median	95% cluster interval
proportional/F2_25	stop_ratio	1.1018	[1.0672, 1.2307]
proportional/F2_25	common_ratio	1.1018	[1.0672, 1.2307]
proportional/F2_25	early_ratio	1.0311	[1.0075, 1.0741]
proportional/F2_25	full_ratio	1.1018	[1.0672, 1.2307]
proportional/F2_50	stop_ratio	1.5052	[1.4078, 1.5928]
proportional/F2_50	common_ratio	1.5052	[1.4078, 1.5935]
proportional/F2_50	early_ratio	0.9813	[0.9286, 1.0276]
proportional/F2_50	full_ratio	1.5052	[1.4078, 1.5962]
relay/F2_25	stop_ratio	1.0996	[1.0331, 1.1568]
relay/F2_25	common_ratio	1.0996	[1.0331, 1.1568]
relay/F2_25	early_ratio	1.0886	[1.0197, 1.1406]
relay/F2_25	full_ratio	1.0996	[1.0331, 1.1568]
relay/F2_50	stop_ratio	1.2567	[1.1152, 1.3345]
relay/F2_50	common_ratio	1.2567	[1.1152, 1.3345]

Law/dose	Estimand	Median	95% cluster interval
relay/F2_50	early_ratio	1.1844	[1.0587, 1.2337]
relay/F2_50	full_ratio	1.2567	[1.1152, 1.3345]

Table S17: Repeated fixed task horizons; these do not add independent samples. No endpoint was selected for significance.

Law	Network	800 ms	900 ms	1000 ms
proportional	healthy	48/48	48/48	48/48
proportional	F2_25	48/48	48/48	46/48
proportional	F2_50	46/48	46/48	37/48
relay	healthy	48/48	48/48	48/48
relay	F2_25	48/48	48/48	48/48
relay	F2_50	48/48	48/48	48/48

Common duration is outcome-dependent and does not hold closed-loop sensory histories fixed. Full-horizon activity after first contact is a specified counterfactual resource measure, not a prospective predictor or a claim about physical contact dynamics. Activity profiles at 200/400/600/800/1000 ms, including counts of prior contacts at each window, are provided in activity-profiles.csv.

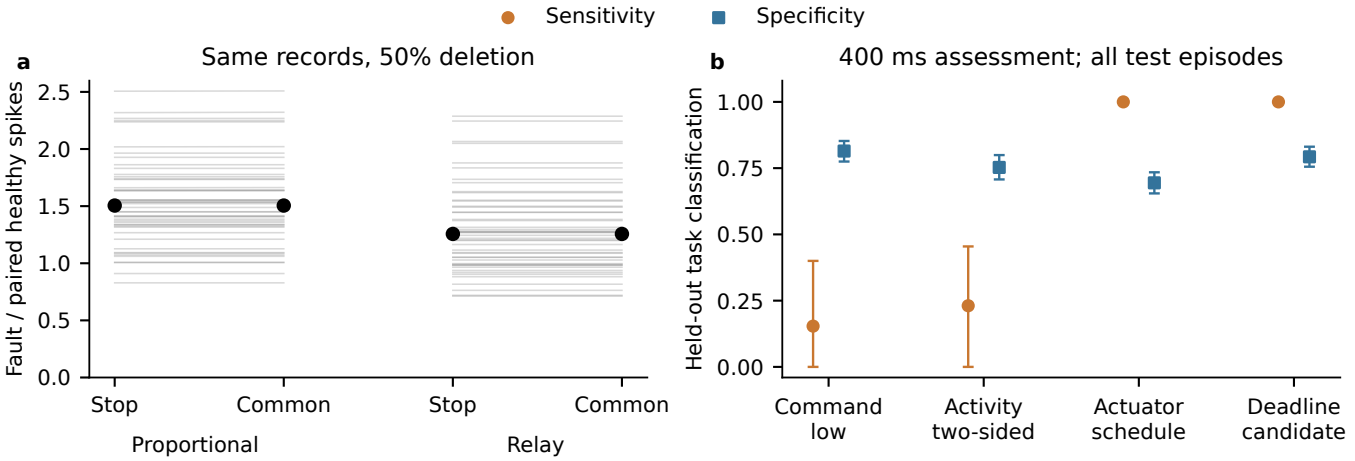


Figure S2: Core exposure estimands and frozen warning performance. Every line in the left panel is a matched scenario; black markers are medians. Right-panel intervals use complete seed-cluster resampling. The upper resource criterion and all subgroup errors remain tabulated even if they are not emphasized in the main figure.

5 Prespecified off-angle stress transfer

The ±45° test uses eight NEW seed clusters and unchanged laws, map, actuator and thresholds. These angles were absent from calibration and validation. The activity reference transfers from the same-side 30° stratum. No threshold is fitted on stress outcomes, and stress records are not pooled with core records to strengthen inference. It tests angle transfer, not a new connectome or RF map. Eight-cluster uncertainty is necessarily wide.

Table S18: Stress outcomes, eight clusters per law/direction/dose. Counts are separate from the 24-cluster core.

Law	Network	ϕ	Success	L	G	H at risk
proportional	healthy	-45	8/8	0	0	8
proportional	healthy	+45	8/8	0	0	8
proportional	F2_25	-45	8/8	0	0	8
proportional	F2_25	+45	8/8	0	0	8
proportional	F2_50	-45	6/8	2	0	8
proportional	F2_50	+45	8/8	0	0	8
relay	healthy	-45	8/8	0	0	8
relay	healthy	+45	8/8	0	0	8
relay	F2_25	-45	8/8	0	0	8
relay	F2_25	+45	8/8	0	0	8
relay	F2_50	-45	8/8	0	0	8
relay	F2_50	+45	8/8	0	0	8

Table S19: Every stress outcome. Definitions match the core outcome table.

Seed	ϕ	P H	P 25	P 50	R H	R 25	R 50
1100	-45	1	1	1	1	1	1
1100	+45	1	1	1	1	1	1
1101	-45	1	1	0	1	1	1
1101	+45	1	1	1	1	1	1
1102	-45	1	1	1	1	1	1
1102	+45	1	1	1	1	1	1
1103	-45	1	1	0	1	1	1
1103	+45	1	1	1	1	1	1
1104	-45	1	1	1	1	1	1
1104	+45	1	1	1	1	1	1
1105	-45	1	1	1	1	1	1
1105	+45	1	1	1	1	1	1
1106	-45	1	1	1	1	1	1
1106	+45	1	1	1	1	1	1
1107	-45	1	1	1	1	1	1
1107	+45	1	1	1	1	1	1

Table S20: Stress warning confusion counts; unchanged calibration and exact simulated geometry. Missing conditional rates remain undefined.

Criterion	Subset	TP	FP	TN	FN	Sens	Spec	FPR	FNR
Command low	all	1	40	54	1	0.500	0.574	0.426	0.500
Command low	faulty	1	33	29	1	0.500	0.468	0.532	0.500
Command low	healthy	0	7	25	0	–	0.781	0.219	–
Own-stop upper	all	0	2	92	2	0.000	0.979	0.021	1.000
Own-stop upper	faulty	0	2	60	2	0.000	0.968	0.032	1.000
Own-stop upper	healthy	0	0	32	0	–	1.000	0.000	–
400-ms upper	all	0	0	94	2	0.000	1.000	0.000	1.000
400-ms upper	faulty	0	0	62	2	0.000	1.000	0.000	1.000
400-ms upper	healthy	0	0	32	0	–	1.000	0.000	–
Two-sided activity	all	0	43	51	2	0.000	0.543	0.457	1.000
Two-sided activity	faulty	0	21	41	2	0.000	0.661	0.339	1.000
Two-sided activity	healthy	0	22	10	0	–	0.312	0.688	–
Actuator schedule	all	2	21	73	0	1.000	0.777	0.223	0.000
Actuator schedule	faulty	2	21	41	0	1.000	0.661	0.339	0.000
Actuator schedule	healthy	0	0	32	0	–	1.000	0.000	–
Held-command candidate	all	2	11	83	0	1.000	0.883	0.117	0.000
Held-command candidate	faulty	2	11	51	0	1.000	0.823	0.177	0.000

Criterion	Subset	TP	FP	TN	FN	Sens	Spec	FPR	FNR
Held-command candidate	healthy	0	0	32	0	–	1.000	0.000	–

The stress set has only two failure clusters. Of 20000 bootstrap draws, 18024 contain a positive class and 1976 contain none. Reported sensitivity intervals condition on the valid draws; the zero observed miss interval is not a finite-sample guarantee. The joint new-cluster-any-miss upper bound is 31.2 percent, not a conditional FNR guarantee. The two-sided activity criterion also flags 22/32 healthy stress episodes: transferring the same-side 30-degree reference does not preserve its healthy flag calibration at 45 degrees.

Table S21: Descriptive stress cluster effects, kept separate from the core analysis.

Seed-level estimand	Mean (pp)	95% interval (pp)
proportional_F2_25_net_loss	0.0	[0.0, 0.0]
proportional_F2_50_net_loss	12.5	[0.0, 31.2]
relay_F2_25_net_loss	0.0	[0.0, 0.0]
relay_F2_50_net_loss	0.0	[0.0, 0.0]
F2_25_interaction	0.0	[0.0, 0.0]
F2_50_interaction	-12.5	[-31.2, 0.0]

6 Separate motor-interface ablations

Baseline walking remains 0.05 m/s when neural actions are disabled; the agent is not immobilized. Burst necessity follows from this toy plant by construction. For a walk-only segment of duration $\Delta t \leq 0.02$ s with precontact separation $d \geq 0.05$ m, the pursuer moves $0.3\Delta t$ toward the segment-start agent position and the agent moves at most $0.05\Delta t$. Since $d > 0.3\Delta t$, triangle inequality gives $d_{\text{next}} \leq d - (0.3 - 0.05)\Delta t$. From $d_0 = 0.25$ m, contact must occur by 0.8 s without a burst, regardless of heading commands. A lightweight geometry property test checks this bound over arbitrary commands. It is not an additional neural experiment or a newly fitted warning metric.

This diagnostic matrix was registered after the original core launch, before complete core analysis. It is not part of the original confirmatory registration and contributes no primary significance test. Eight new seeds 1200–1207 cross the two core bearings with normal proportional/relay, turn-disabled and steering-and-burst-disabled interfaces on healthy networks. The turn-disabled interface retains neural burst triggering; the other removes both neural actions. Common uniforms, calibration and physical settings remain fixed. Every normal failure and every ablated success is retained. The pair of ablations does not separately isolate burst necessity; the kinematic bound above establishes its necessity for this plant. These checks do not establish connectome-specific superiority or biological escape.

Table S22: Independent diagnostic action outcomes. Each law/direction has eight clusters; this cohort is not pooled with core or stress.

Interface	Bearing	Success	Triggered
proportional	-30	8/8	8
proportional	+30	8/8	8
relay	-30	8/8	8
relay	+30	8/8	8
disabled	-30	0/8	8
disabled	+30	0/8	8
observe	-30	0/8	0
observe	+30	0/8	0

Table S23: All action-ablation outcomes, 1=collision-free through 1 s. Turn off retains burst triggering; act off removes both actions.

Seed	Bearing	P	R	Turn off	Act off
1200	-30	1	1	0	0
1200	+30	1	1	0	0
1201	-30	1	1	0	0
1201	+30	1	1	0	0
1202	-30	1	1	0	0
1202	+30	1	1	0	0
1203	-30	1	1	0	0
1203	+30	1	1	0	0
1204	-30	1	1	0	0
1204	+30	1	1	0	0
1205	-30	1	1	0	0
1205	+30	1	1	0	0
1206	-30	1	1	0	0
1206	+30	1	1	0	0
1207	-30	1	1	0	0
1207	+30	1	1	0	0

Table S24: Descriptive normal-minus-ablated success differences (positive means normal succeeds more often), averaging bearings within each of eight clusters. Identical observed contrasts yield degenerate bootstrap intervals, not population certainty.

Seed-level estimand	Mean (pp)	95% interval (pp)
proportional-minus-disabled	100.0	[100.0, 100.0]
proportional-minus-observe	100.0	[100.0, 100.0]
relay-minus-disabled	100.0	[100.0, 100.0]
relay-minus-observe	100.0	[100.0, 100.0]

7 Historical audit and preserved exploratory evidence

These tables describe older protocols and are not independent v9 evidence. The v8 original/pre-cue-gated laws, global Brian2 input seeds and stop-on-collision semantics differ from v9. Its six clusters cannot be pooled with the new laws or called confirmed by a different controller comparison. The original 25% interaction vector [0,0.5,0.5,0,0.5,0] has mean 0.25 and post-hoc symmetric-null p=0.25; the 50% contrast is baseline-driven because both faulty laws have zero survivors. The conditional 25% percentile interval [0.0833,0.4167] does not override the sparse support.

Table S25: Preserved v8 counts independently reconstructed from its 72 records.

Law	Fault	-30	+30	Total
brain	healthy	4	0	4/12
brain	F2_25	4	0	4/12
brain	F2_50	0	0	0/12
precue_gated	healthy	6	0	6/12
precue_gated	F2_25	3	0	3/12
precue_gated	F2_50	0	0	0/12

The source-qualified old-flag audit covers 96 unique records; 6 have an incorrect exact-horizon label. Reused copies are deduplicated. Another 57 old records lack sufficient exact generation-source/timing metadata for this audit; current code is not substituted for that missing provenance. Raw files remain unchanged.

Table S26: Corrected derived v8 flags: all listed intervals end exactly at the 1-s horizon. Their last-frame state is true but exact-horizon state is false.

Law	Fault	Seed	Bearing	Recorded	Exact
brain	F2_25	13	-30	1	0
brain	F2_25	15	-30	1	0
brain	healthy	15	-30	1	0
brain	healthy	17	-30	1	0
precue_gated	F2_25	17	-30	1	0
precue_gated	healthy	13	-30	1	0

Old nominal 250/450-ms actuation is realized as 260-ms delay and 440-ms completed bursts on 20-ms frame starts. Original/gated healthy survivors active at the exact horizon number two/two, versus four/three during the last frame. These metadata corrections do not change recorded survival. New v9 exact-event runs are separately executed and replay-checked; they do not retroactively validate old trajectories.

Table S27: Read-only 600-ms archive recomputation. f is row deletion; p=1-f is survival. No universal power law or recruitment mechanism follows.

f	p	Output retention	Retention/p
0.05	0.95	0.9703	1.0214
0.10	0.90	0.9835	1.0928
0.25	0.75	0.8922	1.1896
0.50	0.50	0.7646	1.5292

Historical transmission-only/joint scoring gives 276/288 versus 262/288 on the SAME records, not runs of SpikeFI or SpikingJET. The 14 differences are 12 inhibitory-hub and two random expected-contact-mass trials (zero-based trials 3 and 9: 327246 and 271058 spikes). Expected contact-mass matching does not equate row fractions (about 8.63% versus 5.57%). Their median does not remove the tail risk. HUB_inh median/mean are 847476.5/847857 spikes. At 75-Hz drive, 50% deletion passes the historical 51.4-Hz criterion in only 10/12 cases; the 150-Hz main result does not extend to all drives.

Table S28: Preserved legacy closed-loop records. Bursts mean any logged burst interval, not exact-horizon activity. Geometry/calibration are compromised; this is a descriptive audit.

Cohort	Survival	Bursts	Above 2x	Median pair ratio
v4/F4_1pct	12/12	12	6	1.7029
v4/F2_25	10/12	10	0	0.8861
v4/F2_50	0/12	0	0	0.3817
v4c/HUB_inh	6/12	12	12	71.6931

F4 has 12/12 survival but 6/12 twofold activity exceedances; its pooled ratio near 1.7 is not a per-condition pass. Legacy 50% deletion changes median ratio from 0.381701 to 0.729443 under common-duration matching. Original F4 node manifests remain unavailable; current reconstructed/protected-FlyID selectors cannot recover those identities or validate the original dynamics. Corrected hub causal experiments are not claimed. The v9 main test uses only row deletion and does not depend on those hub selections.

8 Compute, audit scope and exact artifact access

Table S29: Measured episode wall times. Overlapping worker sums are NOT CPU seconds or campaign elapsed time.

Cohort	Episodes	Median episode (s)	Episode sum (s)
Core	288	145.51	43040.80
Stress	96	133.00	12600.25
Action ablation	64	131.40	8401.72

The exercised runtime is CPython 3.12.14, Brian2 2.10.1, NumPy 2.5.4, Cython 3.1.3 (NumPy code generation), on a Windows CPU host with 31.59 GiB physical memory. Full pins, source/input hashes and actual launcher elapsed/exit-code records accompany the artifact. No hardware mapping, energy, area, physical upset rate or real-time neural execution is inferred. The independent audit checks trace/input consistency using recorded readout outputs; it does not verify every neural state or replace full simulation.

The versioned source/data archive binds the main and supplementary PDFs to TeX, figures, CSV/JSON analysis, every independent raw episode, maps, registrations and exact generating snapshots. External connectivity/completeness tables and dependency caches are excluded; expected input hashes and the upstream release are specified. Reanalysis and input/geometry replay need no external table or new neural execution. Fresh simulation needs the external model and a Git checkout. The public repository may contain an earlier revision; a local archive is not public availability or evidence of submission. See `docs/REPRODUCE_v9.md` and the final material index. Historical source/timestep gaps remain explicit.